UNIVERSITY OF ESSEX

Postgraduate Examinations 2023

PROFESSIONAL PRACTICE AND RESEARCH METHODOLOGY

Time allowed: **TWO** hours

This exam is Open Book (Restricted), students may bring/use the following in the exam:
Study Notes (up to 12 A4 pages)

Candidates are permitted to use:
Casio FX-83GT PLUS/X or Casio FX-85GT PLUS/X only

The paper consists of **NINETEEN** questions. Candidates must answer **ALL** questions. There are 17 multiple choice questions in **SECTION A** and 2 written answer questions in **SECTION B**. **ANSWERS TO SECTION B SHOULD BE TYPED DIRECTLY INTO MOODLE.**

Questions in Section A are of equal weight, questions in Section B are **NOT** of equal weight.

Section A is worth 51% of the paper, Section B is worth 49%.

Please do not leave your seat unless you are given permission by an invigilator.

Do not communicate in any way with any other candidate in the examination room.

If you have a query with the content of this exam paper, or if you have a technical problem with MOODLE, or any other query, please ask an invigilator.

NOTE that you will be marked on your FIRST ATTEMPT only, unless there are technical reasons why you need to start another attempt.

At the end of the examination, remain seated until you have been told you may leave.

Your responses must be your own work. Procedures are in place to detect plagiarism.

SECTION A

SECTION A IS UNAVAILABLE

SECTION B

All ANSWERS to SECTION B must be written in MOODLE

<u>Question 18</u>

The formula for Present Value of money is:

$$\frac{X}{(1+r)^t}$$

Where X is the amount to be adjusted, r is the interest rate as a fraction and t is the time period.

a) A friend needs £500 now and will pay you back £570 in **<u>1</u>** year. Is that a good investment when you can get 10% elsewhere? Please justify your answer with net present value (NPV) calculation. [7%]

b) Will the investment decision change if you can get 15% elsewhere? Please justify your answer with NPV calculation. [7%]

Question 19

a) Define cash flow and explain the importance of cash flow statements? [4%]

b) Two project engineers in a company propose 2 projects. The company needs to take a decision on which project should it invest in. The company raises $3 million cash at 9% interest rate for initial investment (which needs to be returned at 9%) in one of the projects. The cash flow of the 2 proposed projects A & B can be seen in the table below:

Project A		Discounted Cash Flow
Year	Cash Flow	
0	-$3 million, initial investment	…………
1	$2 million profit	…………
2	$4 million profit	…………
3	$4 million profit	…………
4	$2 million profit	…………
5	$0, project closeout	…………

Project B		Discounted Cash Flow
Year	Cash Flow	
0	-$3 million, initial investment	…………
1	$0	…………
2	$0	…………
3	$0	…………
4	$0	…………
5	$ 14 million profit	…………

i) Compute the Discounted Cash Flow for both projects and give your answers for all the years (0-5). [14%]

ii) Calculate the net present value (NPV) of both projects A & B. [8%]

iii) Which project, assuming both carry the same risk, should the financial analyst recommend to the manager? Please explain your reasons in a few sentences. [3%]

iv) Companies have to submit a financial report each year. The following three financial statements are usually the most important sections of the annual report: [6%]

- The Balance Sheet
- The Profit & Loss Account
- The Cash Flow Statement

Describe the purpose of these reports.

END OF SECTION B

END OF PAPER CE902-7-SP